EDUGENIE – PROJECT DEVELOPMENT PHASES

5. Project Development Phase

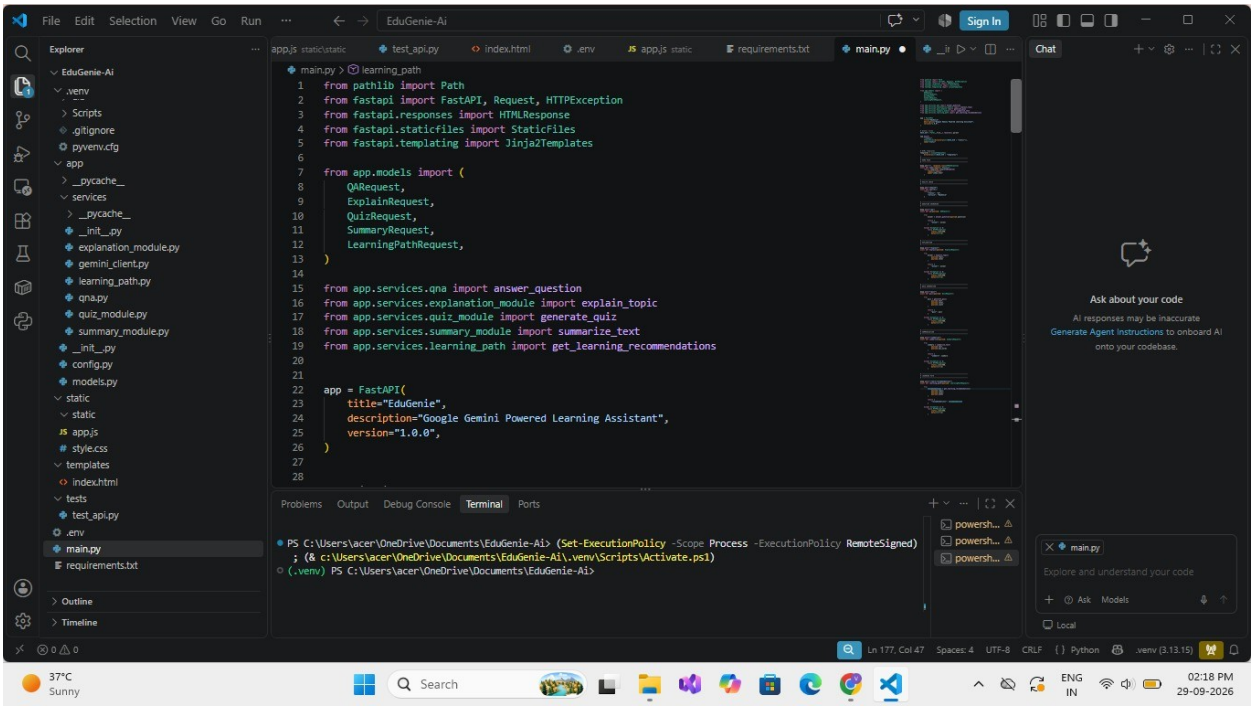
The development phase involves implementing the planned modules and integrating them into a complete application.

5.1 Backend Development

FastAPI was selected as the backend framework because it provides a lightweight and efficient way to create REST APIs.

The backend handles:

- User requests.
- API routing.
- AI model communication.
- Response processing.
- Error handling.

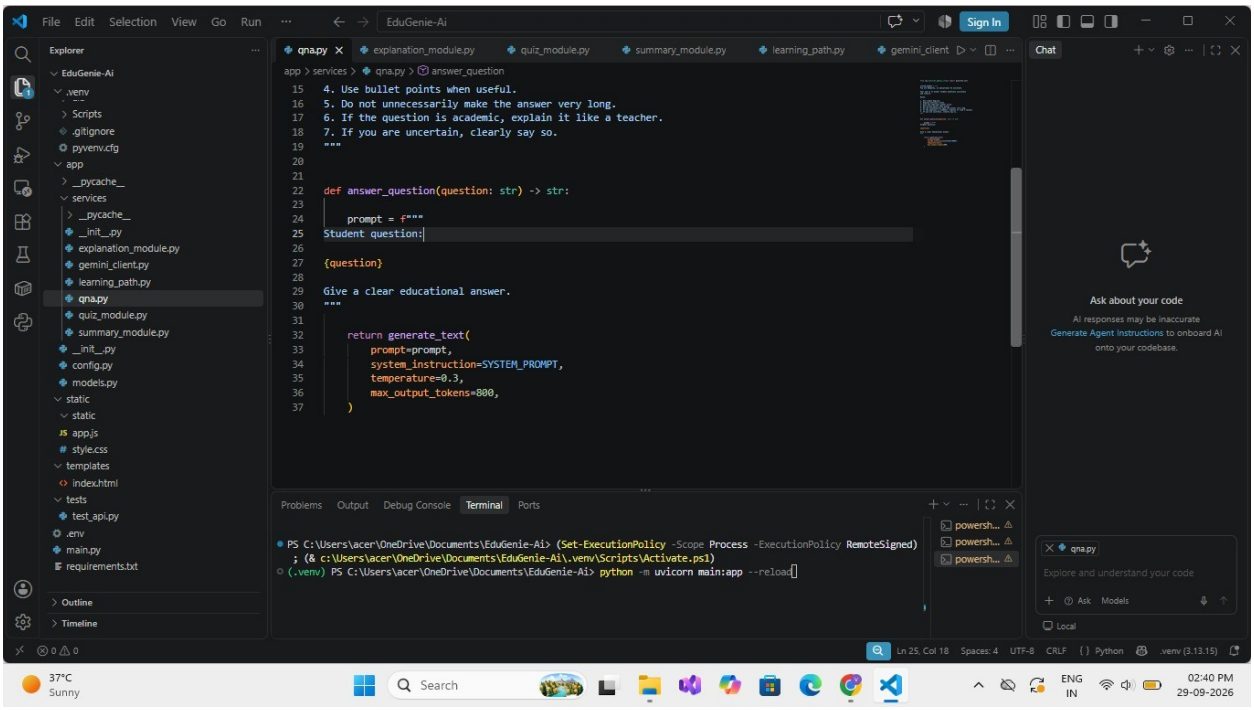


5.2 Q&A Module

The Q&A module accepts a question from the user and sends it to the AI model.

Workflow

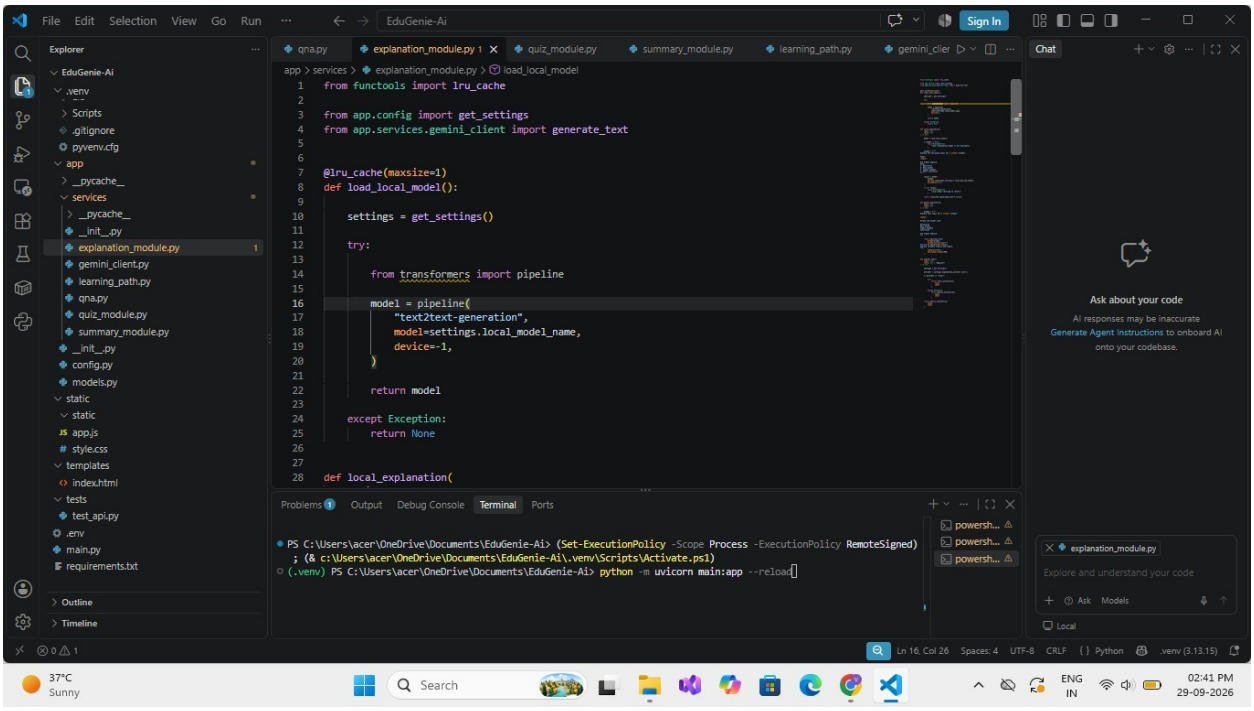
```
User Question
      ↓
FastAPI /qa
      ↓
Q&A Module
      ↓
Gemini API
      ↓
Generated Answer
      ↓
Frontend
```



5.3 Explanation Module

This module converts complex topics into simple explanations suitable for learners.

The lightweight AI model can be used to provide concise educational explanations



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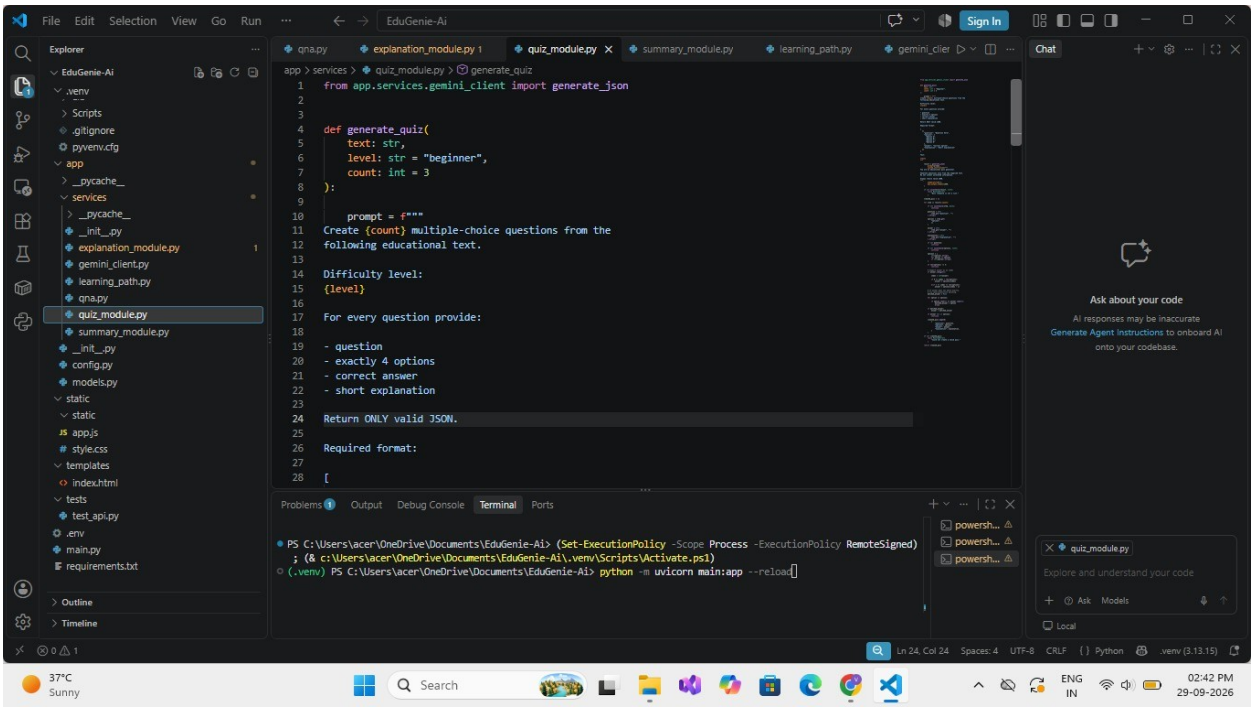
5.4 Quiz Module

The quiz module generates multiple-choice questions.

Each quiz contains:

- Question.
- Four options.
- Correct answer.
- Structured output.

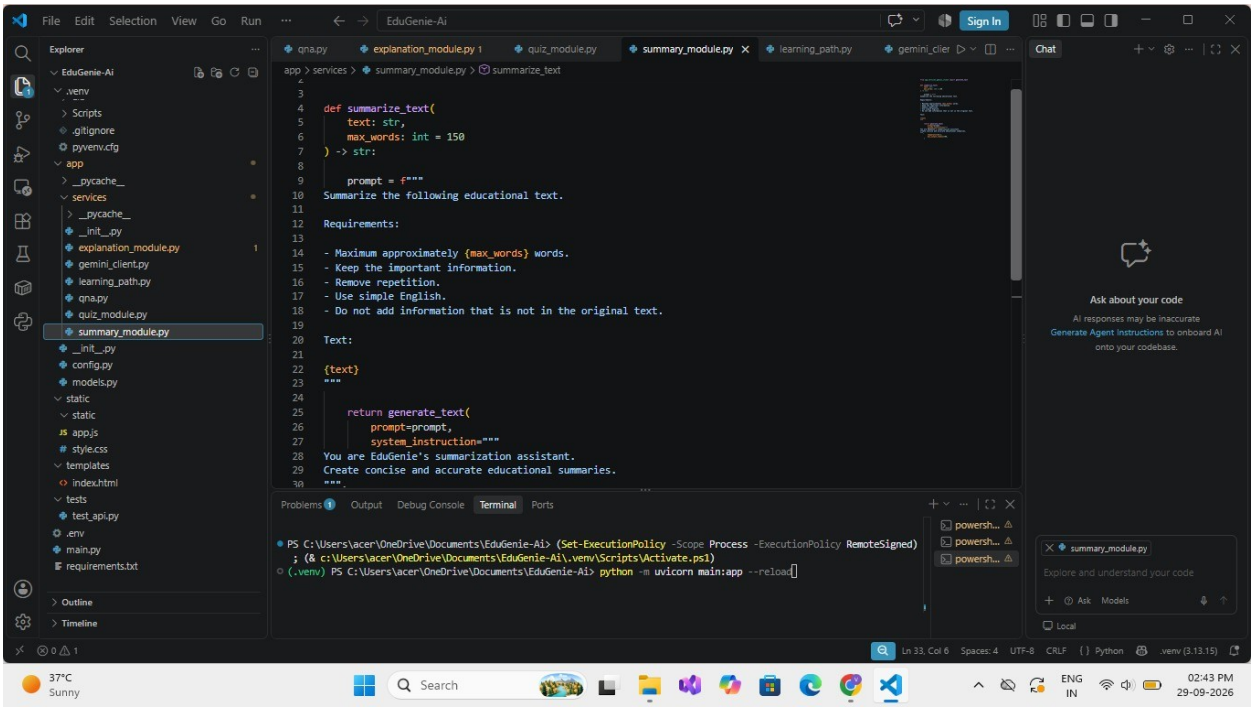
The module also processes the AI response and handles JSON parsing errors.



5.5 Summary Module

The summary module accepts lengthy educational text and generates a concise version.

It removes unnecessary information while retaining the important concepts.

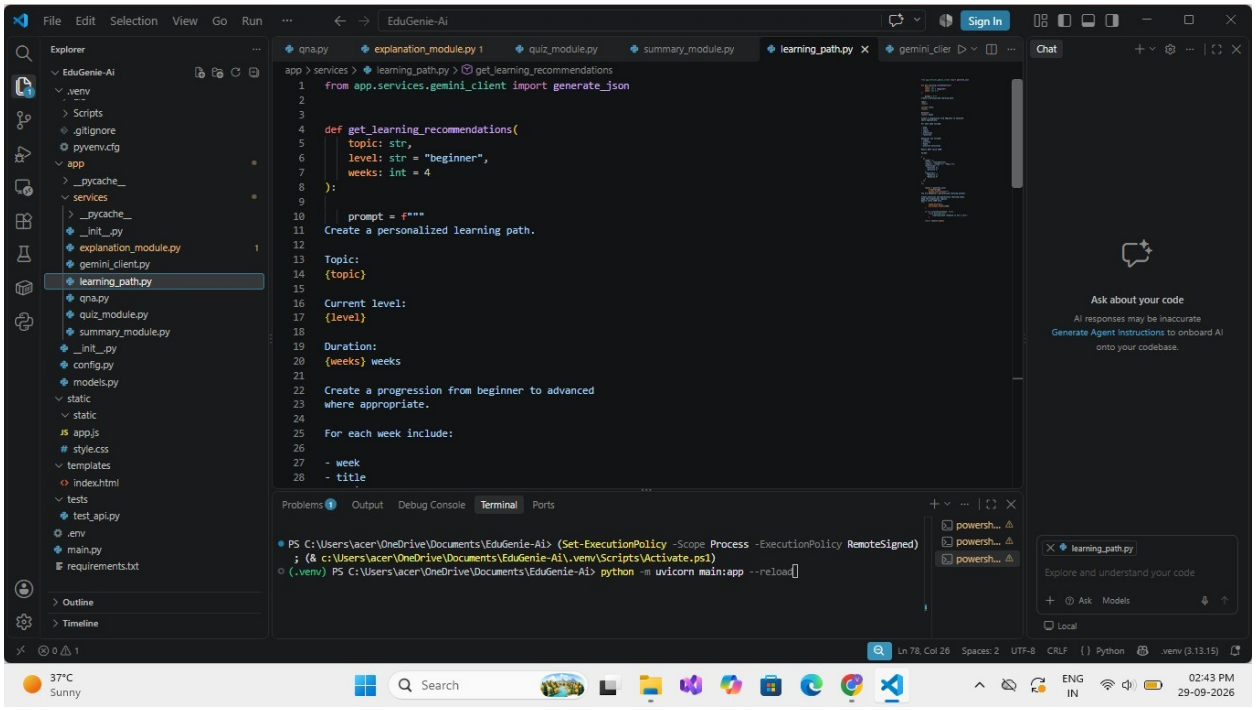


5.6 Learning Path Module

This module generates a personalized learning path.

The learning path can contain:

- Beginner topics.
- Intermediate topics.
- Advanced topics.
- Suggested timeline.
- Learning resources.
- Recommended sequence.



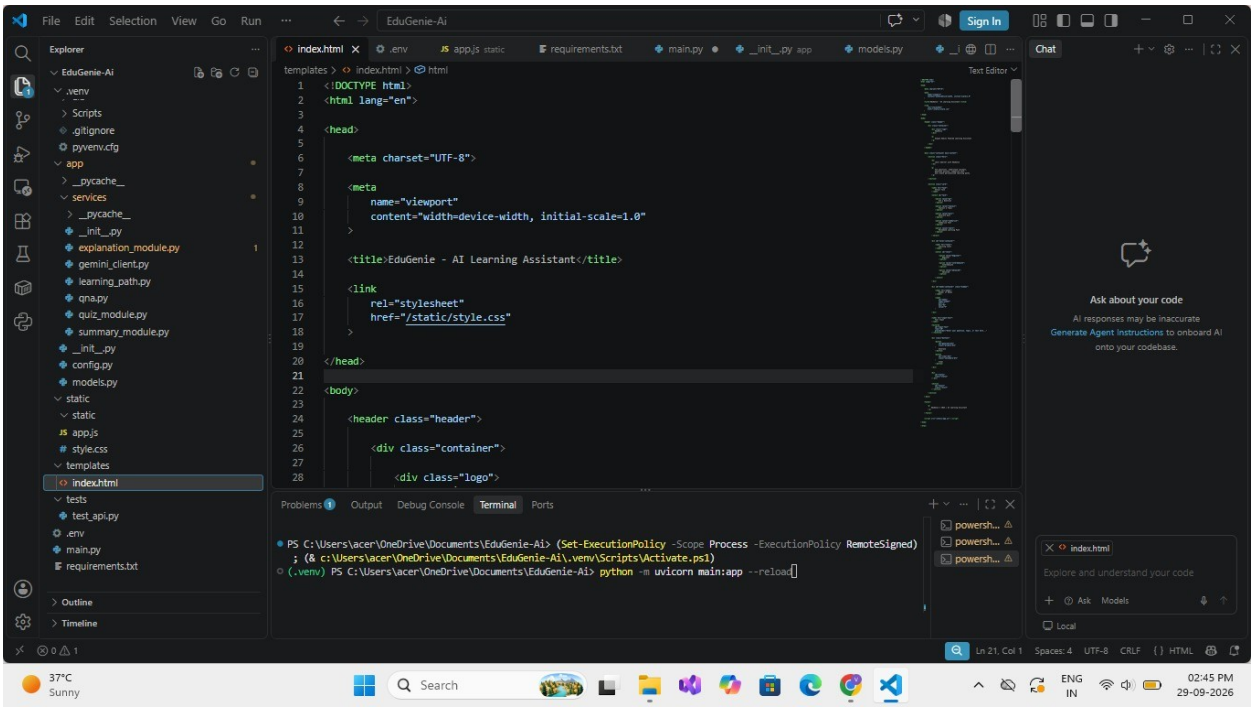
5.7 Frontend Development

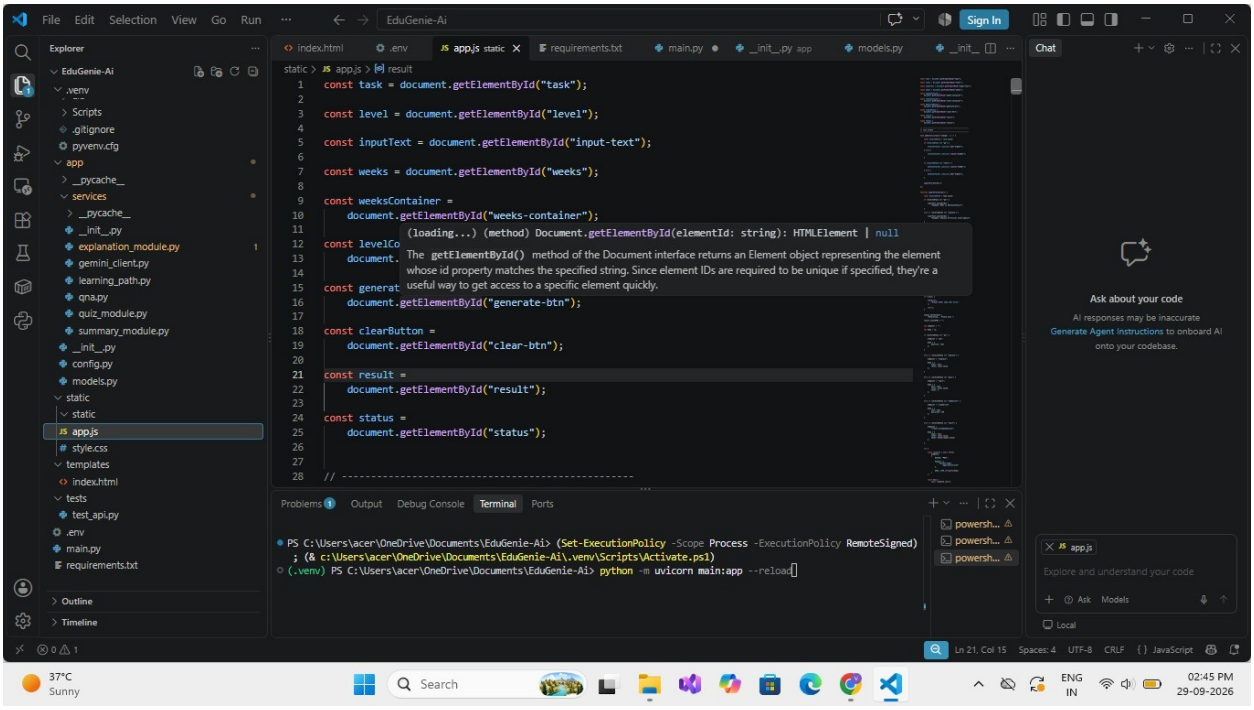
The frontend was developed using HTML and CSS.

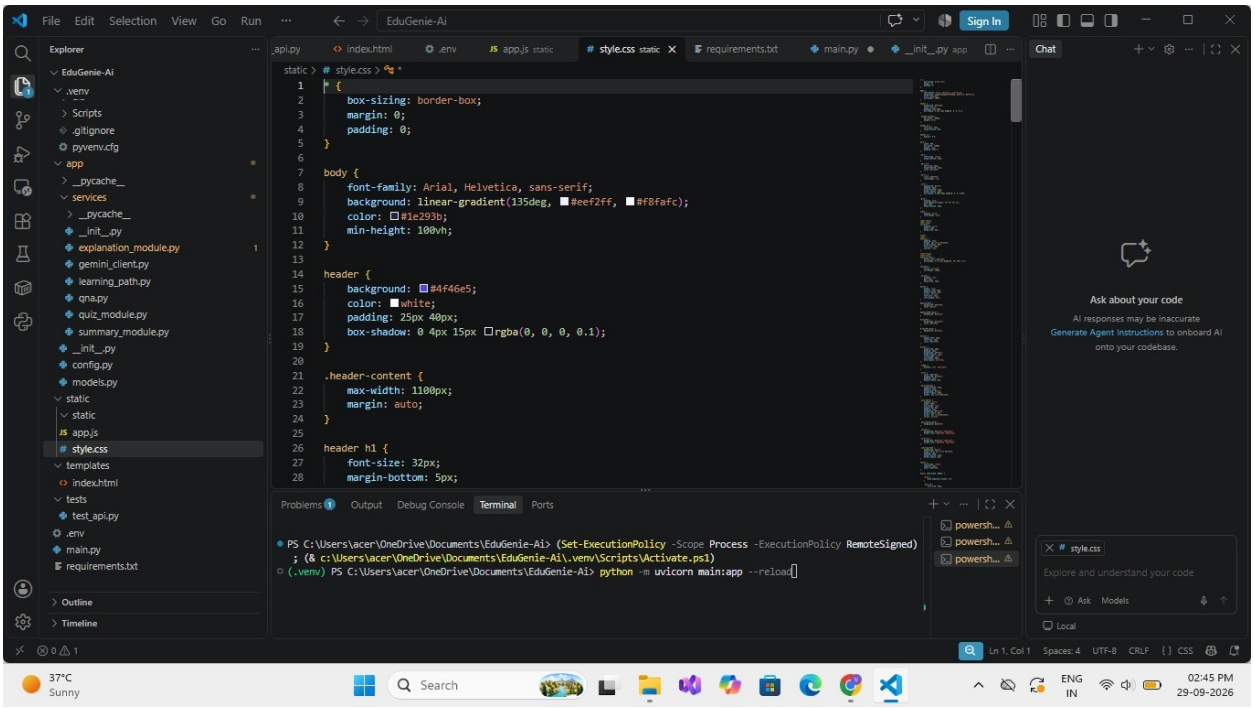
The interface contains:

- Task selection.
- Text input area.
- Submit button.
- Result display section.
- Responsive layout.

The frontend communicates with the FastAPI backend through HTTP requests.






File Edit Selection View Go Run
EduGenie-Ai
Sign In
Explorer
EduGenie-Ai
.venv
Scripts
.gitignore
pyvenv.cfg
app
__pycache__
services
__pycache__
__init__.py
explanation_module.py
gemini_client.py
learning_path.py
qna.py
quiz_module.py
summary_module.py
__init__.py
config.py
models.py
static
static
app.js
style.css
templates
index.html
tests
test_api.py
.env
main.py
requirements.txt
Outline
Timeline
api.py
index.html
.env
app.js static
style.css static
requirements.txt
main.py
__init__.py app
static > style.css > *
* {
box-sizing: border-box;
margin: 0;
padding: 0;
}
body {
font-family: Arial, Helvetica, sans-serif;
background: linear-gradient(135deg, #eef2ff, #f8fafc);
color: #1e293b;
min-height: 100vh;
}
header {
background: #4f46e5;
color: white;
padding: 25px 40px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
}
.header-content {
max-width: 1100px;
margin: auto;
}
header h1 {
font-size: 32px;
margin-bottom: 5px;
Problems 1
Output
Debug Console
Terminal
Ports
PS C:\Users\acer\OneDrive\Documents\EduGenie-Ai> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned)
; (& c:\Users\acer\OneDrive\Documents\EduGenie-Ai\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\acer\OneDrive\Documents\EduGenie-Ai> python -m uvicorn main:app --reload
powersh...
powersh...
powersh...
Chat
Ask about your code
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
style.css
Explore and understand your code
Ask
Models
Local
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
CSS
37°C
Sunny
Search
ENG
IN
02:45 PM
29-09-2026